

Avery Perez

Embedded Systems & IoT Firmware Engineer • 7 Years Experience • MERIQ Index: 85% (Strong Candidate)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: FreeRTOS, ARM Cortex, Embedded Linux, MQTT, I2C/SPI/UART, C/C++, PCB Debugging, Python

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core FreeRTOS service, reducing latency by 25%.
- Led a squad of 5 engineers delivering scalable microservices processing 24M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 67%.

Associate Embedded Systems & IoT Firmware Engineer — *Bosch* (2019 - 2023)

- Designed and maintained ARM Cortex data endpoints with 99% SLA uptime.
- Optimized database indexing and caching layers using Embedded Linux saving \$41k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [FreeRTOS, ARM Cortex, Embedded Linux, MQTT]

Engineered end-to-end production solution leveraging FreeRTOS and ARM Cortex with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [Embedded Linux, MQTT, I2C/SPI/UART, C/C++]

Architected high-resilience system integrating Embedded Linux with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Robotics and Autonomous Systems

University of Waterloo — 2015 - 2019 (Score: 3.55 / 4.0)